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Date: October 2026
Context: Research Proposal Portfolio for Polygence

Classic Shor Algorithm

Version 0.1 Under Correction

$N, p, q, k \in \mathbb{N}$

$r \in (\mathbb{N} \setminus \{0, 1\})$

$N = p \cdot q$ where $\gcd(p, q) = 1$

$$\gcd(N, a) := \begin{cases} d > 1 \Rightarrow N = d \cdot k \Rightarrow a^r \not\equiv 1 (\mathrm{mod}\, d \cdot k) & \text{[Success]} \\ d = 1 \Rightarrow N = d \cdot N \Rightarrow a^r \equiv 1 (\mathrm{mod}\, N) & \text{[Proceed]} \end{cases}$$

$$\boxed{a^r \equiv 1 (\mathrm{mod}\, N)}$$

$$\begin{aligned} r \equiv x (\mathrm{mod}\, 2) &\Rightarrow r = 2q + x \text{ where } q = \lfloor \tfrac{r}{2} \rfloor \in \mathbb{Z} \text{ and } x \in \{0, 1\} \\ &\Rightarrow x = r - 2q \\ &\Rightarrow x := \begin{cases} x = 1 \text{ [Case 1: r is Odd]} \\ x = 0 \text{ [Case 2: r is Even]} \end{cases} \end{aligned}$$

$$\begin{aligned} [\text{Case 1: } x = 1] &\Rightarrow \tfrac{r}{2} \notin \mathbb{N} \\ &\Rightarrow \text{[Failure]} \end{aligned}$$

$$\begin{aligned} [\text{Case 2: } x = 0] &\Rightarrow a^r = \left(a^{\frac{r}{2}}\right)^2 \equiv 1 (\mathrm{mod}\, N) \\ &\Rightarrow \left(a^{\frac{r}{2}}\right)^2 - 1 = \left(a^{\frac{r}{2}} + 1\right)\left(a^{\frac{r}{2}} - 1\right) \equiv 0 (\mathrm{mod}\, N) \\ &\Rightarrow \left(a^{\frac{r}{2}} + 1\right)\left(a^{\frac{r}{2}} - 1\right) = k \cdot N \text{ where } k \in \mathbb{N} \\ &\Rightarrow \text{[Proceed]} \end{aligned}$$

$$\boxed{\left(a^{\frac{r}{2}} + 1\right)\left(a^{\frac{r}{2}} - 1\right) = k \cdot N}$$

$$\gcd\left(a^{\frac{r}{2}} + 1, N\right) := \begin{cases} p = 1 \text{ [Case 2.1: } q = N \text{]} \\ p > 1 \text{ [Case 2.2: } 1 < p < N \text{]} \end{cases}$$

$$\begin{aligned} [\text{Case 2.1: } q = N \text{ }] &\Rightarrow a^{\frac{r}{2}} - 1 = k \cdot N \text{ where } k \in \mathbb{N} \\ &\Rightarrow N = \begin{cases} \gcd\left(a^{\frac{r}{2}} - 1, N\right) \Rightarrow \gcd\left(a^{\frac{r}{2}} + 1, N\right) = 1 \\ \gcd\left(a^{\frac{r}{2}} + 1, N\right) \Rightarrow \gcd\left(a^{\frac{r}{2}} - 1, N\right) = 1 \end{cases} \\ &\Rightarrow \text{[Failure]} \end{aligned}$$

$$
\begin{aligned}
[\text{Case 2.2: } 1 < p < N\ ] &\Rightarrow \gcd\left(a^{\frac{r}{2}} + 1, N\right) = p \text{ where } p > 1 \\
&\Rightarrow \gcd\left(a^{\frac{r}{2}} - 1, N\right) = q \text{ where } q > 1 \\
&\Rightarrow a^{\frac{r}{2}} \not\equiv \pm 1 \\
&\Rightarrow \gcd\left(a^{\frac{r}{2}} + 1, N\right) \cdot \gcd\left(a^{\frac{r}{2}} - 1, N\right) = N \\
&\Rightarrow [\text{Succcess}]
\end{aligned}
$$

$$
\boxed{N = p \cdot q = \gcd\left(a^{\frac{r}{2}} + 1, N\right) \cdot \gcd\left(a^{\frac{r}{2}} - 1, N\right)}
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